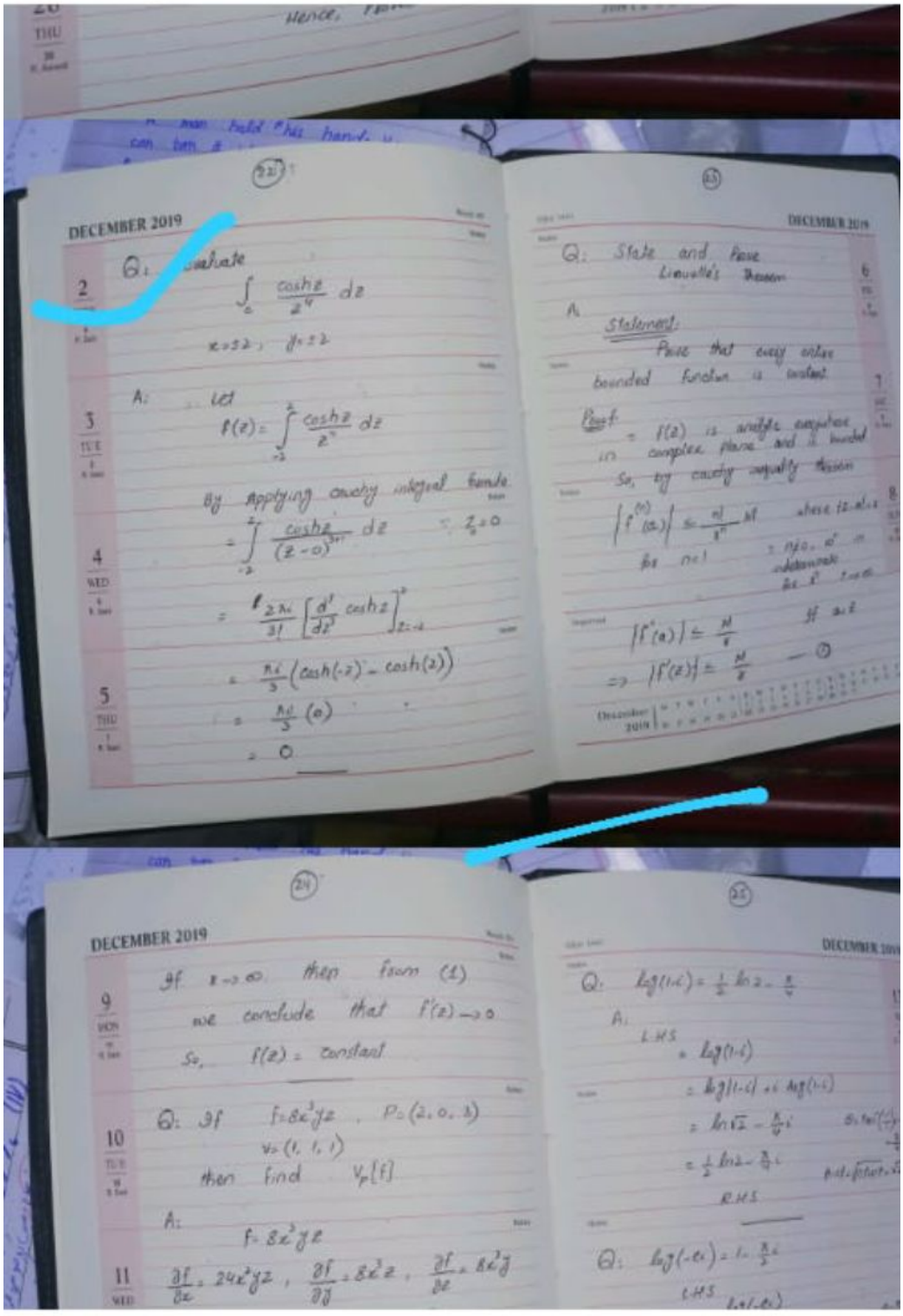
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DECEMBER 2019

Q: Evaluate $\int_C \frac{\cosh z}{z^4}\,dz$ where $C: x=\pm 2,\ y=\pm 2$

A: Let $f(z) = \int_C \frac{\cosh z}{z^n}\,dz$

By applying cauchy integral formula

$= \int_C \frac{\cosh z}{(z-0)^{3+1}}\,dz \quad \because z_0 = 0$

$= \frac{2\pi i}{3!}\left[\frac{d^3}{dz^3}\cosh z\right]_{z=0}$

$= \frac{\pi i}{3}(\cosh(-z) - \cosh(z))$

$= \frac{\pi i}{3}(0)$

$= 0$

Q: State and Prove Liouville's Theorem

A:

Statement:

Prove that every entire bounded function is constant

Proof:

$f(z)$ is entire complex in complex plane and is bounded

So, by cauchy inequality theorem

$|f^{(n)}(z)| \le \frac{n!M}{R^n}$ where $|z-z_0| = R$

for $n=1$

$|f'(a)| \le \frac{M}{R}$

$\Rightarrow |f'(z)| \le \frac{M}{R}$ — (1)

If $R \to \infty$ then from (1) we conclude that $f'(z) \to 0$

So, $f(z) =$ constant

Q: If $f = 8x^2yz$, $P = (2,0,3)$, $v = (1,1,1)$ then find $\nabla_v[f]$

A: $f = 8x^2yz$

$\frac{\partial f}{\partial x} = 24x^2yz$, $\frac{\partial f}{\partial y} = 8x^2z$, $\frac{\partial f}{\partial z} = 8x^2y$

Q: $\log(1+i) = \frac{1}{2}\ln 2 + \frac{\pi}{4}$

A:

LHS

$= \log(1+i)$

$= \log|1+i| + i\arg(1+i)$

$= \ln\sqrt{2} + \frac{\pi}{4}i$

$= \frac{1}{2}\ln 2 + \frac{\pi}{4}i$

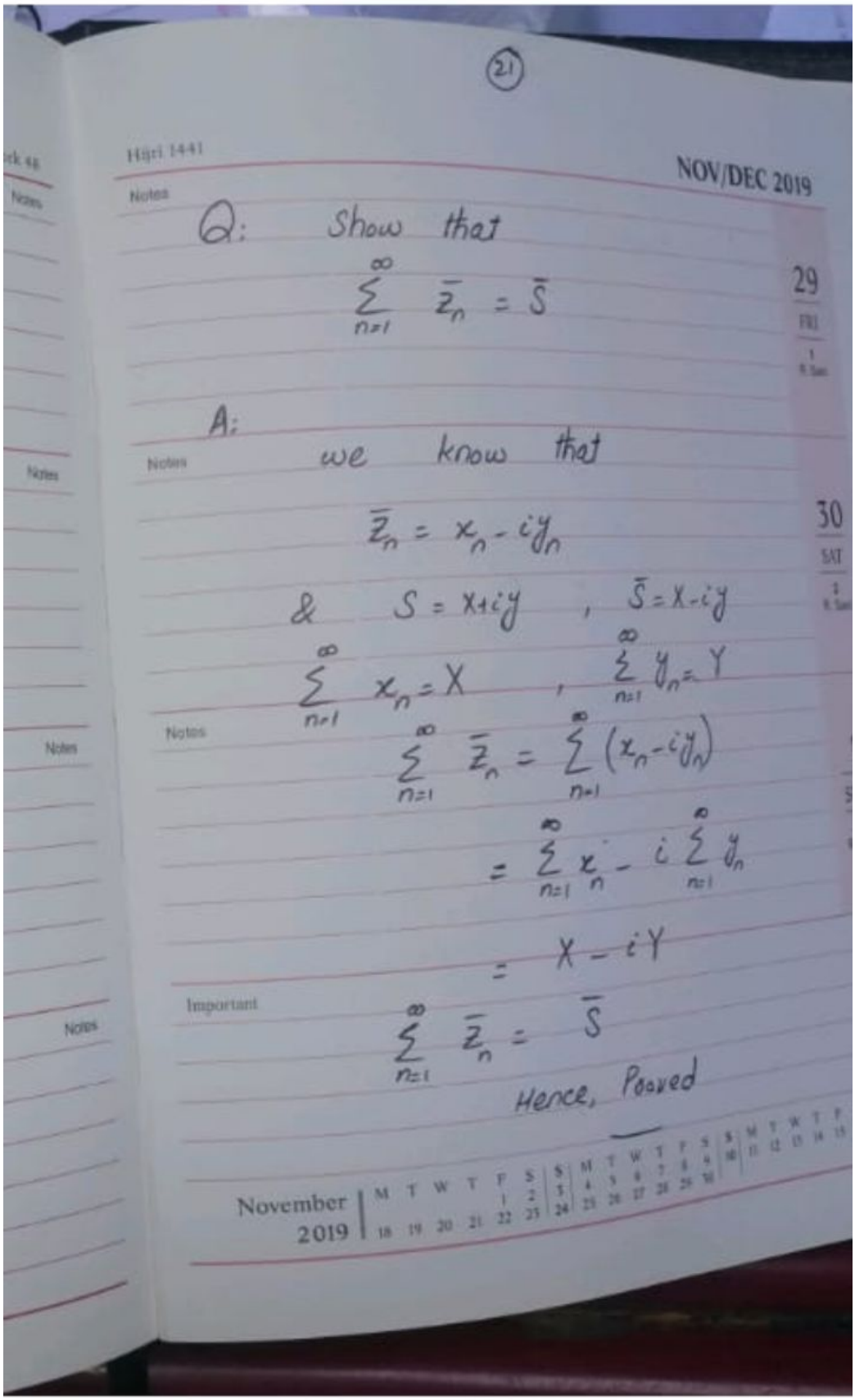
RHS

Q: $\log(-e) = 1 + \frac{\pi}{2}i$

LHS

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(21)

Hijri 1441 — NOV/DEC 2019

Notes

Q: Show that

$$\sum_{n=1}^{\infty} \bar{z}_n = \bar{S}$$

29 FRI

A:

we know that

$$\bar{z}_n = x_n - iy_n$$

30 SAT

& $S = x + iy$, $\bar{S} = x - iy$

$$\sum_{n=1}^{\infty} x_n = X \quad , \quad \sum_{n=1}^{\infty} y_n = Y$$

$$\sum_{n=1}^{\infty} \bar{z}_n = \sum_{n=1}^{\infty} (x_n - iy_n)$$

$$= \sum_{n=1}^{\infty} x_n - i \sum_{n=1}^{\infty} y_n$$

$$= X - iY$$

Important

$$\sum_{n=1}^{\infty} \bar{z}_n = \bar{S}$$

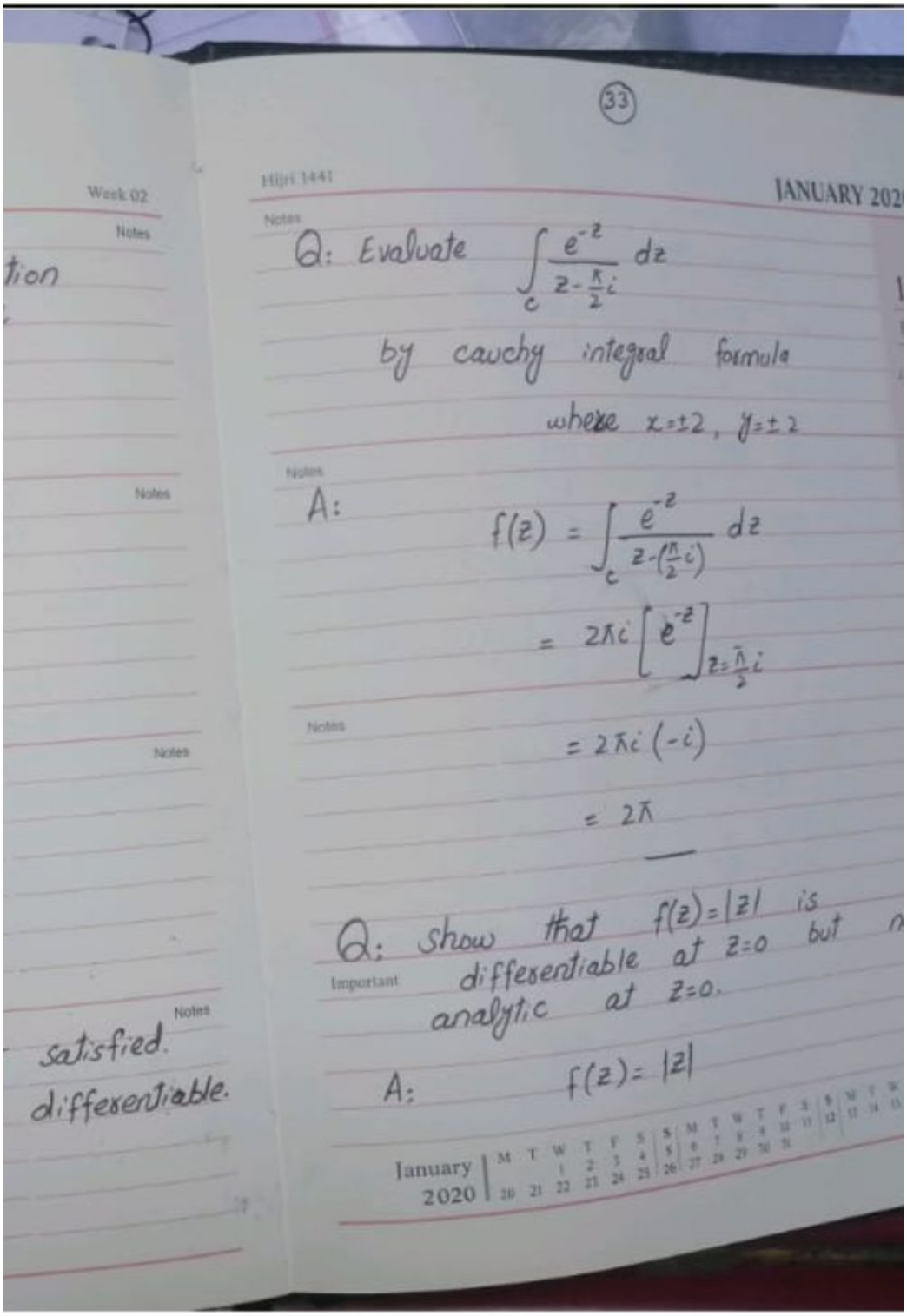
Hence, Proved

November 2019

M	T	W	T	F	S	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

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Week 02

Hijri 1441

JANUARY 202

Notes

tion

Q: Evaluate $\int_C \frac{e^{-z}}{z-\frac{\pi}{2}i}\,dz$

by cauchy integral formula

where $x=\pm 2$, $y=\pm 2$

A:

$$f(z) = \int_C \frac{e^{-z}}{z-(\frac{\pi}{2}i)}\,dz$$

$$= 2\pi i \left[e^{-z}\right]_{z=\frac{\pi}{2}i}$$

$$= 2\pi i\,(-i)$$

$$= 2\pi$$

Q: Show that $f(z)=|z|$ is differentiable at $z=0$ but n analytic at $z=0$.

Important

A: $f(z)=|z|$

satisfied.

differentiable.

January 2020

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$$\lim_{x \to 1} \frac{z^4 - 1}{z^2 + 1}$$

$$\lim_{x \to 1} \frac{(z^2 - 1)(z^2 + 1)}{(z^2 + 1)}$$

$$= \lim_{x \to 1} z^2 - 1$$

Apply $\lim x \to 1$

$$= (1)^2 - 1 = 1 - 1 = 0$$

Evalute Integral $\int_0^{\pi/6} e^{i2t}\, dt$

let

$$I = \int_0^{\pi/6} e^{i2t}\, dt \quad \text{—— (i)}$$

We Know that

$$\frac{d}{dt}\left(e^{i2t}\right) = e^{i2t}(2i)$$

Just we need $(2i)$ in Integral (i)

"$\times$" and "$\div$" Integral (i) by "$2i$"

$$I = \frac{1}{2i}\int_0^{\pi/6} e^{i2t}(2i)\, dt$$

$$= \frac{1}{2i}\left| e^{i2t} \right|_0^{\pi/6}$$

$$I = \frac{1}{2i}\left(e^{i\pi/3} - 1\right) \quad \underline{\text{Ans}}$$

$$\int_C \frac{e^{2z}}{z^3} dz$$

As e^{2z} is analytic at $z=0$ and it lies inside the curve so we will apply Cauchy Integral formula.

$$\int_C \frac{e^{2z}}{z^3} dz = 2\pi i f(0)$$

$$= 2\pi i (e^{2(0)})$$

$$= 2\pi i (e^0)$$

$$= 2\pi i (1)$$

$$= 2\pi i$$

$$\because \int_c f(z)\,dz = \int_a^b f(z(t))\,z'(t)\,dt \longrightarrow (i)$$

Here ~~for z~~

$$z(t) = e^{it}$$

$$z'(t) = ie^{it}$$

$$\because f(z) = \frac{1}{z}$$

$$f(z(t)) = \frac{1}{e^{it}} = e^{-it}$$

$$(i) \Rightarrow \int_c f(z)\,dz = \int_0^{2\pi} e^{-it}(ie^{it})\,dt$$

$$= i\int_0^{2\pi} 1\,dt$$

$$= i(2\pi - 0) = 2\pi i.$$

W T F S S M T W T F S S M T W T F S S M T
6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26

$$\lim_{z \to i} \frac{z^2+4z+2}{z+1}$$

$$= \frac{(i)^2+4i+2}{i+1}$$

$$= \frac{-1+4i+2}{i+1}$$

$$= \frac{1+4i}{i+1}$$

$$= \frac{1+4i}{1+i} \times \frac{1-i}{1-i}$$

$$= \frac{(1-i)+4i-4i^2}{(4)^2-(i)^2}$$

$$= \frac{1-i+4i+4}{1+1}$$

$$= \frac{5+3i}{2} \quad \text{dun}$$

$$\int_0^{\pi/6} e^{i2t}\,dt$$

$$\int_0^{\pi/6} (\cos 2t + i\sin 2t)\,dt$$

$$\int_0^{\pi/6} \cos 2t\,dt + i\int_0^{\pi/6} \sin 2t\,dt$$

$$+\frac{\sin 2t}{2}\Big|_0^{\pi/6} + i\,\frac{-\cos 2t}{2}\Big|_0^{\pi/6}$$

$$\frac{1}{2}\left[\sin 2t\Big|_0^{\pi/6} - i\cos 2t\Big|_0^{\pi/6}\right]$$

$$= \frac{1}{2}\left[\sin\frac{2\pi}{6} - \sin 2(0) - i\left(\cos 2\left(\frac{\pi}{6}\right) - i\cos 0\right)\right]$$

$$= \frac{1}{2}\left[\frac{\sqrt{3}}{2} - 0 - i\,\frac{1}{2} - i\right]$$

$$= \frac{1}{2}\left[\frac{\sqrt{3}}{2} - \left(\frac{i}{2} + i\right)\right]$$

$$= \frac{1}{2}\left[\frac{\sqrt{3}}{2} - \left(\frac{i+2i}{2}\right)\right]$$

$$= \frac{1}{2}\left[\frac{\sqrt{3}}{2} - \frac{3i}{2}\right]$$

$$= \frac{\sqrt{3}}{4} - \frac{3i}{4}$$

$$\int_0^{1+i} f(z)\,dz$$

$f(z) = z$ is given by

$$\int_0^{1+i} z\,dz$$

$$\frac{z^2}{2}\Big|_0^{1+i} = \frac{1}{2}\left[z^2\right]_0^{1+i}$$

$$= \frac{1}{2}\left[(1+i)^2 - (0)^2\right]$$

$$= 1/2\left(-(1+i^2-2i)\right)$$

$$= 1/2\left(-(1+(-1)-2i)\right)$$

$$= 1/2\left(-1+1+2i\right)$$

$$1/2\,(2i) = i \text{ Ans}$$

MTH 632

$$\frac{d}{dz}\arcsin z = \frac{1}{\sqrt{1-z^2}}$$

$$\frac{d}{dz}\arccos z = \frac{-1}{\sqrt{1-z^2}}$$

$$\frac{d}{dz}\arctan z = \frac{1}{1+z^2}$$

$$\operatorname{arcsinh} z = \log\left[z + (z^2+1)^{1/2}\right]$$

$$\operatorname{arccosh} z = \log\left[z + (z^2-1)^{1/2}\right]$$

$$\operatorname{arctanh} z = \frac{1}{2}\log\left(\frac{1+z}{1-z}\right)$$

$$\arcsin z = -i\log\left(iz + (1-z^2)^{1/2}\right)$$

$$\arccos z = -i\log\left(z + (1-z^2)^{1/2}\right)$$

$$\arctan z = \frac{i}{2}\log\left(\frac{i+z}{i-z}\right)$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

$$\cos z = \frac{e^{iz} + e^{iz}}{2}$$

$$\sinh z = \frac{e^{z} - e^{-z}}{2}$$

$$\cosh z = \frac{e^{z} + e^{z}}{2}$$

OVEMBER 2019

Q: Prove that

$$\frac{d}{dz}(\sinh^{-1} z) = \frac{1}{\sqrt{1+z^2}}$$

A: Let

$w = \sinh^{-1} z$

$z = \sinh w$

Taking derivative

$$\frac{d}{dz}(z) = \frac{d}{dz}(\sinh w)$$

$$1 = \frac{d}{dw}(\sinh w) \cdot \frac{dw}{dz}$$

$$1 = -\cosh w \cdot \frac{dw}{dz}$$

$$-\frac{1}{\cosh w} = \frac{dw}{dz}$$

$$\frac{dw}{dz} = \frac{1}{\sqrt{1 + (\cosh w)^2}}$$

$$\frac{d}{dz}(\sinh^{-1} z) = \frac{1}{\sqrt{1+z^2}}$$

Q: Evaluate

$$\int_C \frac{\cosh z}{z^4}\, dz$$

$x = \pm 2, \quad y = \pm 2$

A: Let

$$f(z) = \int_{-2}^{2} \frac{\cosh z}{z^4}\, dz$$

By Applying cauchy integral formula

$$= \int_{-2}^{2} \frac{\cosh z}{(z-0)^{3+1}}\, dz \qquad \because z_0 = 0$$

$$= \frac{2\pi i}{3!} \left[\frac{d^3}{dz^3} \cosh z \right]_{z=-2}^{2}$$

$$= \frac{\pi i}{3} \left(\cosh(-2) - \cosh(2) \right)$$

$$= \frac{\pi i}{3} (0)$$

$$= 0$$

$$\int_C \frac{\cosh z}{z^4}\,dz$$ by cauchy integral formula.

Solution:

We know the cauch Integral formula:

$$\int_C \frac{f(z)\,dz}{(z-z_0)^{n+1}} = \frac{2\pi i}{n!} f^{(n)}(z_0)$$

∴ function is analytic

Show that $f(z) = e^z$ is analytic

Sol: $f(z) = e^z$

$u + iv = e^{x+iy}$

$= e^x \cdot e^{iy}$

$= e^x (\cos y + i \sin y)$

(2 8).

$u + iv = e^x \cos y + i e^x \sin y$

$u = e^x \cos y \quad , \quad v = e^x \sin y$

$u_x = e^x \cos y \quad , \quad v_x = e^x \sin y$

$u_y = -e^x \sin y \quad , \quad v_y = e^x \cos y$

Here

$u_x = v_y$ and $u_y = -v_x$

So function is analytic

Given an example such that u and v are harmonic but $u+iv$ is not analytic

Solution

Consider the function

$f(z) = z$

$u + iv = x + iy$

$u = x \quad , \quad v = y$

$u_x = 1 \quad , \quad v_x = 0$

$u_y = 0 \quad , \quad v_y = 1$

(27)

Test the analyticity of function $w = \sin z$

Sol:

Let $w = f(z)$

$f(z) = \sin z$

$u + iv = \sin(x + iy)$

$= \sin x \cosh y + i \cos x \sinh y$

Now

$u = \sin x \cosh y \; ; \; v = \cos x \sinh y$

$u_x = \cos x \cosh y \; ; \; v_x = -\sin x \sinh y$

$u_y = \sin x \sinh y \; ; \; v_y = \cos x \cosh y$

$u_x = v_y$ and $u_y = -v_x$

$\therefore$ function is analytic

Show that $f(z) = e^z$ is analytic

Sol: $f(z) = e^z$

$u + iv = e^{x+iy}$

$= e^x \cdot e^{iy}$

$= e^x (\cos y + i \sin y)$

(3)

$$\int_0^\infty e^{-zt}\,dt$$

$$\frac{e^{-zt}}{-z}\Big|_0^\infty$$

$$= \frac{1}{-z}\left[e^{-zt}\right]_0^\infty$$

$$= \frac{1}{-z}\left[e^{-z(0)} - e^{-z(\infty)}\right]$$

$$= 1/_{-z}\left[e^{0} - e^{\infty}\right]$$

$$= 1/_{-z}\,(1)$$

$$= 1/_{-z}$$

Imp

Question: $z_n = \frac{1}{n^3} - i$ solve it for convergence also find value of ε

Solution:

$$z_n = \frac{1}{n^3} - i$$

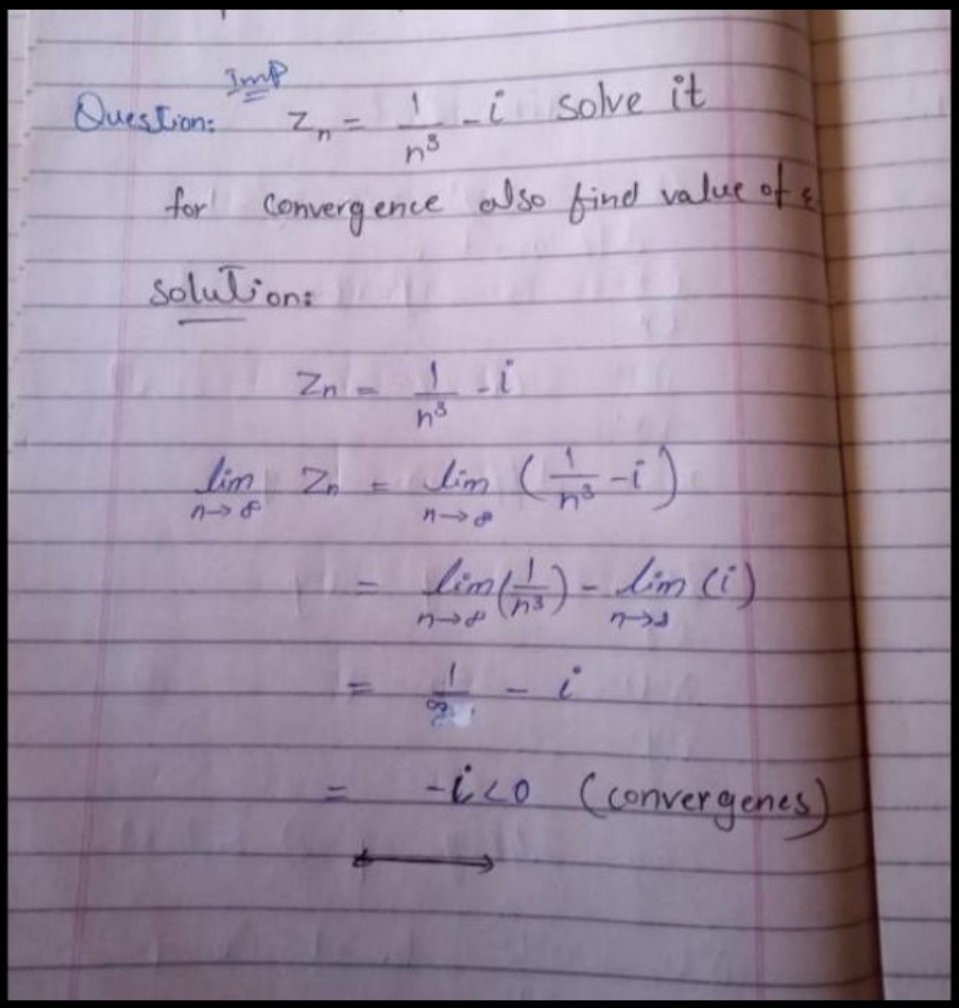
$$\lim_{n \to \infty} z_n = \lim_{n \to \infty} \left(\frac{1}{n^3} - i\right)$$

$$= \lim_{n \to \infty} \left(\frac{1}{n^3}\right) - \lim_{n \to \infty} (i)$$

$$= \frac{1}{\infty} - i$$

$$= -i < 0 \text{ (convergenes)}$$

11:08 83



Imp
Question: $z_n = \frac{1}{n^3} - i$ solve it for convergence also find value of ε

Solution:

$$z_n = \frac{1}{n^3} - i$$

$$\lim_{n\to\infty} z_n = \lim_{n\to\infty}\left(\frac{1}{n^3} - i\right)$$

$$= \lim_{n\to\infty}\left(\frac{1}{n^3}\right) - \lim_{n\to\infty}(i)$$

$$= \frac{1}{\infty} - i$$

$$= -i < 0 \text{ (convergenes)}$$

Tel/Mob/Fax/E-mail

Name & Address

Q: $z_n = -2 + i\frac{(-1)^n}{n^2}$ $(n=1,2,\ldots)$

$\lim_{n\to\infty} z_n$ converges or not?

A:

$$\lim_{n\to\infty} z_n = \lim_{n\to\infty}\left(-2 + i\frac{(-1)^n}{n^2}\right)$$

$$= \lim_{n\to\infty}(-2) + \lim_{n\to\infty}\left[i\frac{(-1)^n}{n^2}\right]$$

$$= -2 + i\lim_{n\to\infty}\frac{(-1)^n}{n^2}$$

$$= -2 + i(0)$$

$$= -2 < 0$$

So, z_n converges.

Name & Address

Tel/Mob/Fax/E-mail

Q: $\int_0^\infty e^{-zt}\,dt \qquad Re\,z > 0$

A:

$$I = \int_0^\infty e^{-zt}\,dt$$

$$= \lim_{b\to\infty} \int_0^b e^{-zt}\,dt$$

$$= \lim_{b\to\infty} \left[\frac{e^{-zt}}{-z}\right]_0^b$$

$$= -\frac{1}{z} \lim_{b\to\infty} \left(e^{-zb} - e^{-0b}\right)$$

$$= -\frac{1}{z} \lim_{b\to\infty} \left(e^{-zb} - e^{0}\right)$$

$$= -\frac{1}{z}(0-1)$$

$$= \frac{1}{z} \quad \text{when} \quad Re\,z > 0$$

———

JANUARY 2020

Week 04

20 MON

Q: Prove that

$$v = \frac{1}{2}\log(x^2+y^2)$$

A:

$$v(x,y) = \frac{1}{2}\log(x^2+y^2)$$

21 TUE

$$\frac{\partial v}{\partial x} = \frac{x}{x^2+y^2} \quad , \quad \frac{\partial v}{\partial y} = \frac{y}{x^2+y^2}$$

$$\frac{\partial^2 v}{\partial x^2} = \frac{-x^2+y^2}{(x^2+y^2)^2} \quad , \quad \frac{\partial^2 v}{\partial y^2} = \frac{x^2-y^2}{(x^2+y^2)^2}$$

22 WED

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = -\frac{x^2-y^2}{(x^2+y^2)^2} + \frac{x^2-y^2}{(x^2+y^2)^2}$$

$$= 0$$

Function is Harmonic

Now, to find v

23 THU

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} = \frac{x}{x^2+y^2}$$

By integrating w.r.t y

$$v(x,y) = \tan^{-1}\left(\frac{y}{x}\right) + g(x)$$

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Taking derivative w.r.t x

$$\frac{\partial v}{\partial x} = -\frac{y}{x^2+y^2} + g'(x)$$

But $\frac{\partial v}{\partial x} = -\frac{y}{x^2+y^2}$

$$-\frac{y}{x^2+y^2} = -\frac{y}{x^2+y^2} + g'(x)$$

$$g'(x) = 0$$

$$g(x) = c$$

$$v(x,y) = \tan^{-1}\left(\frac{y}{x}\right) + c$$

Q: Find the arc length

$$S = \int_a^b \left|\frac{d\vec{x}}{dt}\right| dt$$

17

$$\vec{x}(t) = (\cos t, \sin t, t)$$

where $0 \le t \le 2\pi$

A:

18

$$\vec{x}(t) = (\cos t, \sin t, t)$$

$$|\vec{x}'(t)| = |-\sin t, \cos t, 1|$$

$$= \sqrt{2}$$

19

$$\int_0^{2\pi} |\vec{x}'(t)|\, dt = \int_0^{2\pi} \sqrt{2}\, dt$$

$$= \sqrt{2} \int_0^{2\pi} dt$$

$$= \sqrt{2}(t)\Big|_0^{2\pi} = \sqrt{2}(2\pi - 0)$$

$$= \sqrt{2}(2\pi) = 2\sqrt{2}\pi$$

$$= \frac{2\pi}{3!}(0)$$

$$= 0$$

$u(x,y) = 2x(1-y)$ harmonic in some domain and find a harmonic conjugate $v(x,y)$ when

$$u(x,y) = 2x(1-y)$$

Solution:

It is straightforward to show that $u_{xx} + u_{yy} = 0$